

SWATHI K N

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SUMMARY

Computer Science graduate skilled in Java, Spring Boot, Node.js, and Python, with practical experience across REST APIs, microservices, and cloud deployment (AWS, Docker, CI/CD). Independent projects spanning backend systems and LangChain, RAG-based AI tooling. Seeking SDE-1 roles in backend, full-stack, or AI-driven engineering.

TECHNICAL SKILLS

Languages: C++, Java, Python, JavaScript, TypeScript, SQL

Backend: Spring Boot, Node.js, Express.js, FastAPI, REST APIs, Microservices, Kafka, JWT

Databases & Caching: PostgreSQL, Redis, FAISS

Cloud & DevOps: AWS (EC2, S3, Lambda), Docker, GitHub Actions, CI/CD

Tools & Practices: Git, Agile/Scrum, Unit Testing, OpenAPI/Swagger

Concepts: Data Structures and Algorithms (DSA), System Design, OOP, DBMS, Operating Systems, Computer Networks, Problem Solving

AI / ML: LangChain, LangGraph, RAG, Ollama

Certifications: Crash Course on Python by Google (Score: 83.12%), Artificial Intelligence Fundamentals - IBM SkillsBuild

EXPERIENCE

Student Trainee

Jan 2026 - Apr 2026

Rakuten India

Bengaluru, India

- Built and delivered 3+ full-stack applications across frontend and backend, contributing to 10+ features spanning API integration and core application functionality.
- Developed secure web solutions with JWT-based authentication, protected routes, and role-based access control supporting 3+ user roles.
- Implemented backend systems using Node.js, Express.js, PostgreSQL, and Spring Boot, building 20+ REST API endpoints across 4+ relational entities (Users, Projects, Tasks, Comments).
- Deployed and automated workflows using 5 AWS services (EC2, S3, Lambda, CloudWatch, VPC/IAM), Docker, and GitHub Actions, building a 3-stage CI/CD pipeline for automated build and deployment.
- Built 3 AI-driven applications (trip planner agent, Notion automation agent, PDF-based RAG system) using LLMs, FAISS, and embeddings, and backend systems integrating Kafka and Redis for event-driven microservices.

PROJECTS

Multi-Model Evaluation Dashboard | *FastAPI, Ollama, Python, SQLite*

- Built a multi-model evaluation dashboard comparing latency, token usage, and response quality across 3 LLM configurations (Llama 3, Mistral, Phi-3) on identical prompts
- Implemented LLM-as-judge scoring against a fixed 1-5 rubric, automatically grading each model's response and surfacing quality scores alongside performance metrics
- Observed up to 3x latency variance (25s-76s) and identified a length bias in LLM-as-judge scoring through manual review of results across multiple test prompts
- Stored evaluation run history in SQLite, enabling side-by-side comparison across prompts over time via a FastAPI-served dashboard

URL Shortener Service (Bitly-like) | *Spring Boot, Java, PostgreSQL, Redis*

- Designed a scalable URL shortening service using 6-character Base62 encoding (56.8 billion possible codes), with collision-safe retry logic and click analytics supporting horizontal scaling
- Built 5 layered REST API endpoints (Controller, Service, Repository) including shorten, redirect, stats, and paginated listing, with IP-based rate limiting capped at 5 requests/minute via Bucket4j
- Implemented Base62 short-code generation with database-backed collision detection and link expiry validation
- Added centralized exception handling and Redis-backed caching to improve backend reliability

Personal Finance Tracker | *Node.js, Express, TypeScript, PostgreSQL, Redis*

- Developed a backend personal finance tracker with 8 REST endpoints across 3 entities (User, Entries, DailySummary) using Node.js, Express.js, TypeScript, TypeORM, and PostgreSQL
- Implemented JWT authentication with access/refresh token rotation and bcrypt password hashing (10 salt rounds), plus 4 tiered rate limiters (20-100 requests per window) to prevent abuse
- Built filtering, sorting (3 allow-listed fields), and pagination (default 10/page) with Redis caching (20s TTL, auto-invalidated on write) to reduce repeated DB queries
- Automated financial summary generation via a scheduled cron job running every 10 minutes, computing per-user income, expense, and balance totals

Next-Gen Waste Management: Smart Sensor-Based System | *IoT, Arduino, Supabase*

- Designed an IoT-based smart waste management system with real-time bin monitoring, collaborating with operations teams to optimize workflows
- Built sensor-enabled bins using Arduino (IR, proximity, environmental sensors) for intelligent waste detection
- Developed a full-stack web app with React, Tailwind CSS, and Supabase, presenting role-based dashboards to executive leadership (Admin, Driver, User)
- Integrated ThingSpeak for real-time data visualization and optimized waste collection responsiveness

LEADERSHIP & ACTIVITIES

- **Coordinator**, IT Virtuoso Student Forum, Dept. of CSE, Global Academy of Technology, Bengaluru
- **Organizing Committee Member**, IET India Scholarship Award 2024 - Regional Rounds (South 2)
- Progressed from Volunteer (Feb 2024) to Coordinator for Hack-a-League 3.0 (Feb 2025) and 4.0 (Feb 2026) - 24-hour offline hackathons, Dept. of CSE, Global Academy of Technology, Bengaluru
- **Volunteer**, Youth For Seva - week-long educational camp for government school children

EDUCATION

Global Academy of Technology	Bengaluru, Karnataka
<i>Bachelor of Engineering in Computer Science CGPA: 8.75/10</i>	<i>2022 - June 2026</i>
Surana Ind PU College	Bengaluru, Karnataka
<i>Pre-University (PCMC)</i>	<i>2020 - 2022</i>